LAB 4- github link

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CS3807 – Deep Learning Laboratory

Experiment 4

Comparative Study of Deep Convolutional Neural Network Architectures Using Transfer Learning

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1. Objective

The objectives of this experiment are

- Study the evolution of deep CNN architectures.
- Compare LeNet-5, AlexNet, VGG16, GoogleNet and ResNet.
- Understand transfer learning.
- Fine tune pretrained CNN models.
- Compare classification performance of different architectures.

2. Learning Outcomes

After completing this experiment students will be able to

1. Explain modern CNN architectures.
2. Differentiate LeNet, AlexNet, VGG16, GoogleNet and ResNet.
3. Implement transfer learning.
4. Fine tune pretrained CNN models.
5. Compare CNN architectures using performance metrics.

3. Introduction

Deep Convolutional Neural Networks have become the dominant models for computer vision applications because they automatically learn hierarchical features from images. The evolution of CNNs has focused on improving classification accuracy while reducing computational complexity and training difficulties.

The progression of CNN architectures began with LeNet-5 and has evolved through AlexNet, VGGNet, GoogleNet, and ResNet. Each architecture introduced important innovations that significantly improved image classification performance.

4. Evolution of CNN Architectures

Model	Year	Major Contribution
LeNet-5	1998	First practical convolutional neural network
AlexNet	2012	ReLU activation, Dropout and GPU training
VGG16	2014	Uniform 3×3 convolution filters
GoogleNet	2014	Inception modules for multi-scale feature extraction
ResNet	2015	Residual learning using skip connections

5. LeNet-5

LeNet-5 was proposed by Yann LeCun in 1998 for handwritten digit recognition. It consists of two convolution layers, two average pooling layers and fully connected layers. It demonstrated that convolutional networks could automatically learn image features without handcrafted feature extraction.

Advantages

- Small number of parameters
- Fast training
- Suitable for OCR

6. AlexNet

AlexNet won the ImageNet Challenge in 2012 by reducing the classification error significantly. It introduced ReLU activation, Dropout regularization and GPU training.

Major innovations

- ReLU activation
- Dropout
- GPU implementation
- Data augmentation

7. VGG16

VGG16 demonstrated that using multiple small 3×3 convolution filters could outperform larger filters while increasing network depth.

Advantages

- Excellent feature extraction
- Simple architecture
- High classification accuracy

8. Comparison of Architectures

Architecture	Depth	Parameters	Main Contribution
LeNet-5	7	60K	First CNN
AlexNet	8	61M	ReLU + Dropout
VGG16	16	138M	Deep 3×3 filters
GoogleNet	22	6.8M	Inception Modules
ResNet50	50	25.6M	Residual Learning

9. GoogleNet (Inception Network)

GoogleNet, proposed by Szegedy *et al.* in 2014, introduced the **Inception Module**, which performs multiple convolution operations in parallel using different filter sizes. Instead of using only a single kernel size, GoogleNet simultaneously applies 1×1, 3×3, 5×5 convolutions and max pooling. The outputs are concatenated to obtain multi-scale feature representations.

The use of 1×1 convolutions before larger kernels reduces the number of trainable parameters and computational complexity.

Advantages

- Multi-scale feature extraction.
- Fewer parameters than VGG16.
- High computational efficiency.
- Better classification performance.

10. ResNet

Increasing CNN depth often causes the **vanishing gradient problem**, making optimization difficult. ResNet solves this issue using **skip (residual) connections**, allowing gradients to flow directly through the network.

Instead of learning

$$H(x)$$

ResNet learns

$$F(x) = H(x) - x$$

thus,

$$Output = F(x) + x$$

where

- $F(x)$ = Residual Mapping
- x = Identity Mapping

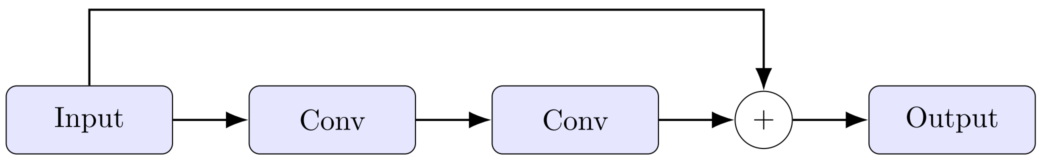

Figure 1: Residual Block in ResNet

Advantages

- Eliminates vanishing gradients.
- Enables very deep neural networks.
- Faster convergence.
- High ImageNet accuracy.

11. Dilated Convolution

Dilated convolution (also called atrous convolution) increases the receptive field without increasing the number of parameters.

The dilation rate determines the spacing between kernel elements.

For a standard convolution,

$$D = 1$$

For dilated convolution,

$$D > 1$$

Example

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 4 & 0 & 5 \end{bmatrix}$$

where zeros indicate inserted gaps.
Applications

- Semantic segmentation
- Medical image analysis
- Satellite imagery
- Object localization

12. Transpose Convolution

Transpose convolution increases the spatial dimensions of feature maps. Unlike pooling, which reduces image size, transpose convolution performs learnable upsampling.

Applications

- Image Super Resolution
- Autoencoders
- GANs
- Image Generation
- Semantic Segmentation

13. Transfer Learning

Transfer Learning utilizes a pretrained CNN model that has already learned general image features from a large dataset such as ImageNet.

Instead of training from scratch,

1. Load a pretrained model.
2. Remove the original classifier.
3. Freeze convolution layers.
4. Add new Dense layers.
5. Train the classifier.
6. Fine tune selected convolution layers.

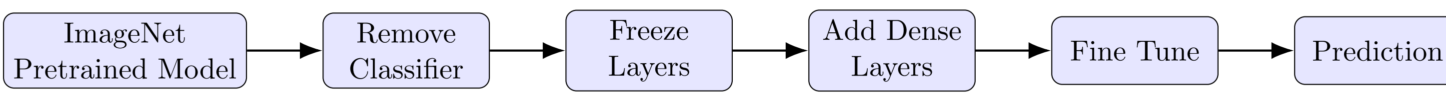

Figure 2: Transfer Learning Workflow

Advantages

- Faster convergence.
- Higher accuracy on small datasets.
- Reduced training time.
- Lower computational cost.

14. Dataset

Dataset: CIFAR-10

- Training Images : 50,000
- Testing Images : 10,000
- Number of Classes : 10
- Image Size : $32 \times 32 \times 3$
- Classes: Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship and Truck.

15. Experimental Procedure

Task 1: Dataset Preparation

1. Load the CIFAR-10 dataset using TensorFlow/Keras.
2. Normalize the pixel values to the range [0,1].
3. Display ten sample images.
4. Print the dimensions of the training and testing datasets.

Task 2: Transfer Learning

Load any one of the following pretrained CNN models.

- VGG16
- ResNet50
- MobileNetV2
- EfficientNetB0 (Optional)

Perform the following steps.

1. Load pretrained ImageNet weights.
2. Remove the original classification layer.
3. Freeze the convolutional base.
4. Add a Global Average Pooling layer.
5. Add a Dense layer with ReLU activation.
6. Add the output layer with Softmax activation.

Task 3: Model Training

Train the model using

Optimizer	Adam
Learning Rate	0.001
Batch Size	32
Epochs	10–20
Loss Function	Categorical Cross Entropy
Evaluation Metric	Accuracy

Task 4: Fine Tuning

1. Unfreeze the last convolution block.
2. Train for another 5–10 epochs.
3. Compare the classification accuracy before and after fine tuning.

Task 5: Model Evaluation

Evaluate the trained model using

- Accuracy

- Precision
- Recall
- F1-score
- Confusion Matrix
- Classification Report

16. Hyperparameter Study

Compare the performance using different settings.

Hyperparameter	Values
Learning Rate	0.001, 0.0001
Batch Size	16, 32, 64
Epochs	10, 20
Optimizer	Adam, SGD
Dense Units	128, 256
Frozen Layers	All, Partial

17. Mandatory Plots

17.1 Sample CIFAR-10 Images

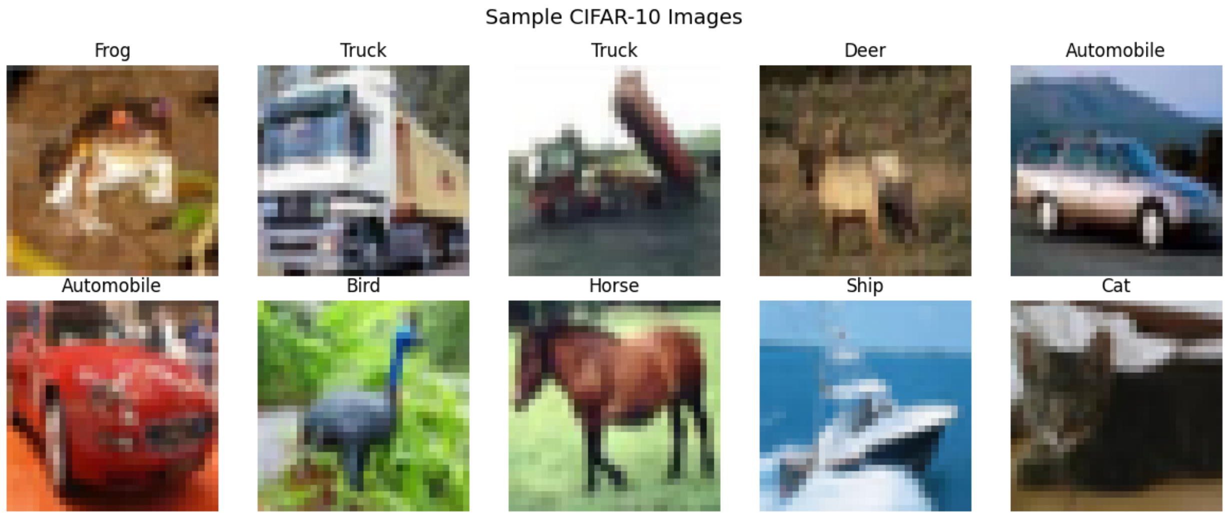

Figure 3: Sample images from the CIFAR-10 dataset

Inference: The samples in the above image show the different CIFAR-10 classes.

17.2 Training and Validation Accuracy

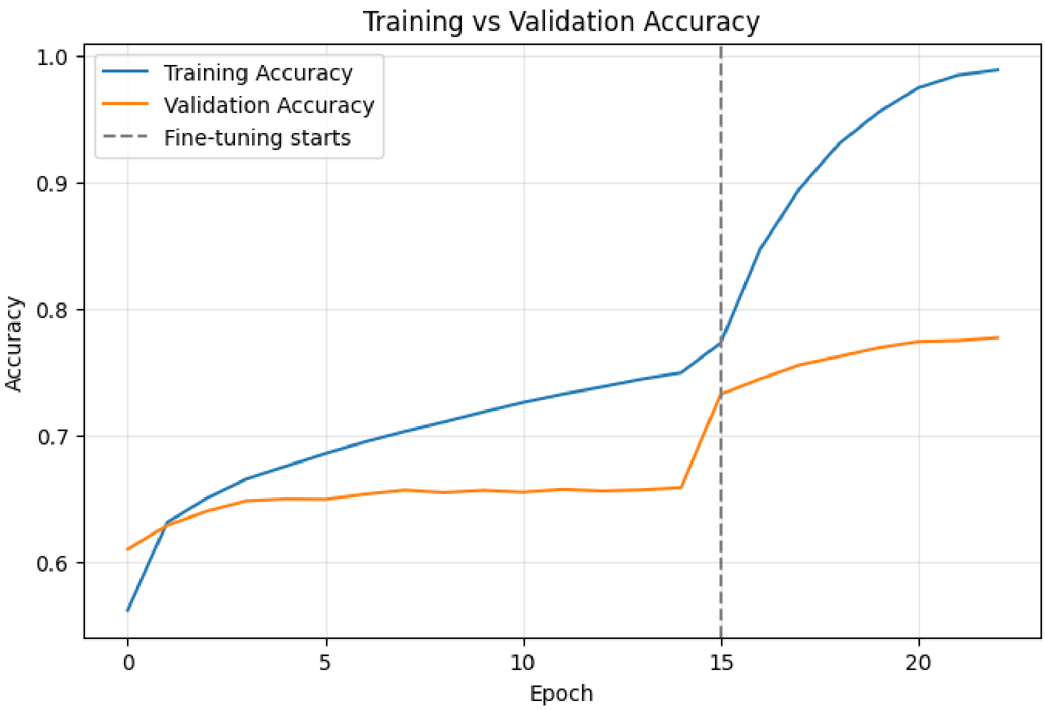

Figure 4: Training and validation accuracy across epochs

Inference: Accuracy improves markedly after fine tuning begins at epoch 15. Training accuracy reaches about 99%, while validation accuracy levels off near 78%, indicating some overfitting.

17.3 Training and Validation Loss

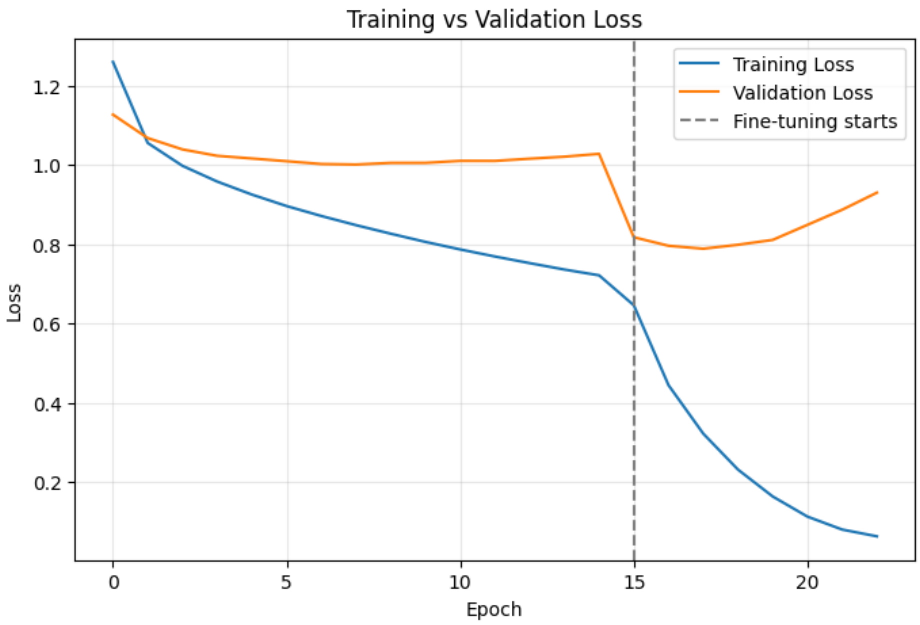

Figure 5: Training and validation loss across epochs

Inference: Fine tuning causes a sharp reduction in both losses initially. The later rise in validation loss, despite continuously falling training loss, suggests increasing overfitting.

17.4 Confusion Matrix

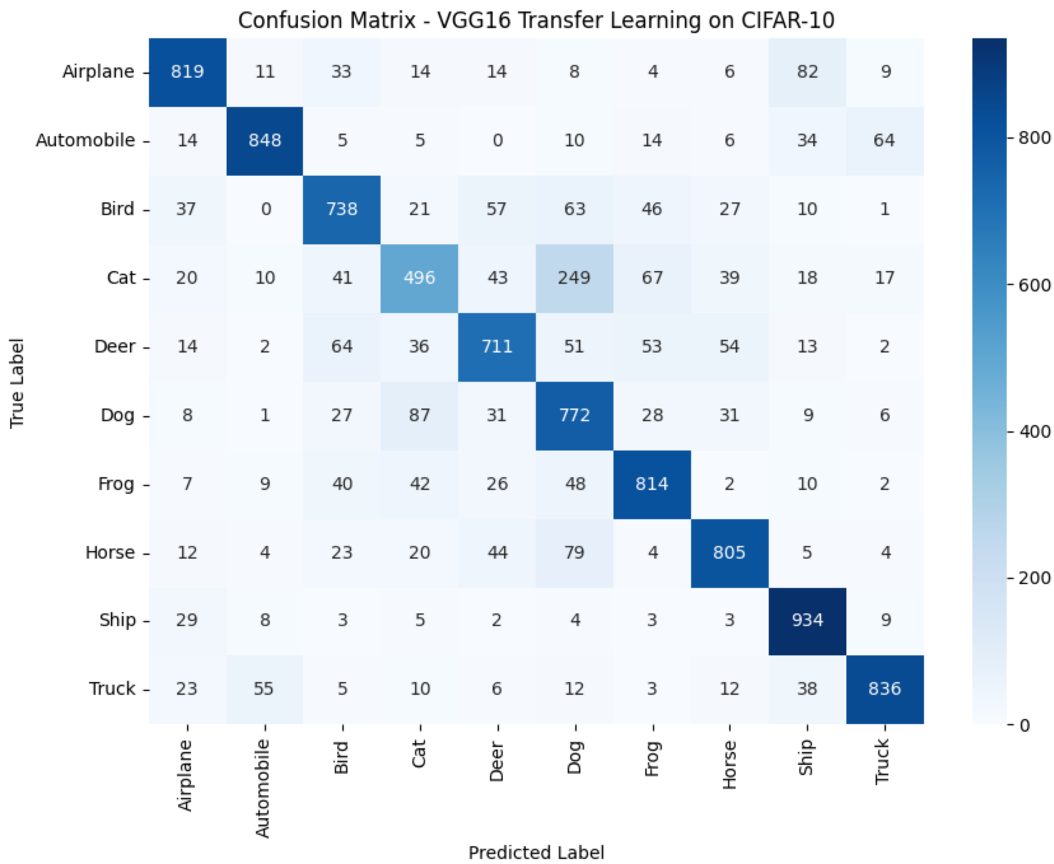

Figure 6: Confusion matrix for VGG16 transfer learning on CIFAR-10

Inference: The model performs best on the ship class and also classifies automobiles and trucks reliably. The largest confusion is between cats and dogs, which have similar visual features at low resolution.

17.5 Misclassified Images

Figure 7: Examples of images misclassified by the trained model

Inference: Many errors involve visually similar classes or images with ambiguous shapes and backgrounds. The examples illustrate the difficulty of recognizing objects in 32×32 pixel images.

18. Results

18.1 Performance Metrics

Metric	Value
Training Accuracy	98.86%
Testing Accuracy	77.73%
Precision	0.7805
Recall	0.7773
F1-score	0.7757
Training Time	1086.6 s
Total Parameters	14,781,642

18.2 Comparison of CNN Architectures

Model	Parameters	Accuracy (%)	Training Time (s)
LeNet-5	60K	-	-
AlexNet	61M	-	-
VGG16	14,781,642	77.73	1086.6
GoogleNet	6.8M	-	-
ResNet50	23,851,274	41.28	390.3

19. Discussion Questions

Answer the following questions.

1. Why is AlexNet considered a breakthrough in deep learning?

   AlexNet was the first CNN to win by a large margin, as it proved that deep CNN trained on gpu can outperform hand-crafted features, it also introduced reLU and prqctical GPU training, ehich made the large-scaled deep learning feasible

2. Why does VGG16 use only 3×3 convolution filters? Using mtiple 3x3 layers gives the same

   receptive field as a largers filter which using fewer parameters and also adding non-linear activation functions, which improves the model's discriminative power whoithout increasing computational resources.

3. Explain the advantages of the Inception module. The inception modeule applies multiple filter

   sizes and pools in parallel and add the outputs, which in turn captures the multi-scale features in a single layer. The 1x1 convolutions before large filters reduce channel depth first, cutting paramters and resources but also making sure accuracy is hugh

4. What is the purpose of residual learning? residual layer allows the layer to learn the residual

   function instead of a full mapping. this give the gradients a direct path in the network to work with, which solves the vanishing gradient problem, furthermore, this helps to make it possible to train much deeper models and also converge faster

5. Differentiate LeNet and ResNet. LeNet is a shallow layered neural network with around 60,000

   parameters that hasbeen deisgned for small size image datasets, which uses aveage pooling and no skip connection, meanwhile RestNet is a 50 layered network with aroun 25 million parameters that uses residual connection to train very deep architectures on large scale , high resolution coloured images and acheiving higher accuracy

6. What is Transfer Learning? transfer learning is where a model which has already been pretrained (mostly generic parts of an image) is being resued as a feature extractor instead of learning everything from scratch which can be used by freezing/reusing the learned CNN features and training only specific layers related to the tasks.
7. Why is fine tuning required? fine tuning is requires as it unfreezes some pre trained layers and continues training with low learning rate, which allows the model to adapt tge generic imageNet features to learn t the new dataset, which helps in improving the accuracy
8. Explain the difference between dilated convolution and transpose convolution. The difference between dilated convolution and transpose convolution is that dilated covilution adds gaps between kernels to make the reveptive field bigger without adding paramter, which is useful for tasks like segmentation, transpose convolution performs something called learnable unsampling, increasing the size of the feature map, which is mainly used for generation or reconstruction tasks
9. Why do pretrained models converge faster? The reason that the pretrained models converge faster is because the weights are already encoded for general purpose of any image learning process like learning edges, textures, shapes which it would have learnt form millions of images,. Thus it only needs to learn and update for the new specific task laters and require less no of epocjs to converge as it does not need to learn from scratch
10. Compare the computational complexity of LeNet and ResNet. From the above table in 18.2, we can see that leNet-5 has around 60,000 parameter and a few shallow layers, making it light and fast to train and run even on CPU, RestNet50 has a huge number of parameters, roughly aroun 26.5 million with 50 layers which would definitely require high GPU resouces, but this also helps in better accuracy and can easily deal with complex image datasets

20. Additional Exercises

1. Implement transfer learning using VGG16.

Model	Parameters	Accuracy (%)	Training Time (s)
VGG16	14,781,642	77.73	1086.6

2. Repeat the experiment using ResNet50.

Model	Parameters	Accuracy (%)	Training Time (s)
ResNet50	23,851,274	41.28	390.3

3. Compare Adam and SGD optimizers.

   Adam and SGD are both optimization algorithms used to update the models weights during training. Adam automatically adjusts the learning rate for each parameter, so it usually learns faster and gives better results with less tuning. In your experiment, Adam achieved 65.01percent validation accuracy, while **SGD achieved only 59.19 percent. Therefore, for this model and dataset, Adam performed better than SGD, although SGD can sometimes work well with careful learning-rate tuning.

4. Compare training with frozen layers and fine tuning.

   Frozen layers mean that the layers in the pretrained models are left as it is without updating anything and training only the final layers for the training. Fine tuning on the other hand unfreezes a few layers and updates them to the wights depending on the specific task

5. Evaluate the model using Precision, Recall and F1-score.

   Based on the performace metrics in 18.1, where the model's Precision is 0.7805, Recall is 0.7773, and F1-score is 0.7757 which shoows that the model performed averagely well as most of the vaues are between 0.7-0.8 which show good generalizaton

6. Compare the performance of LeNet, AlexNet and ResNet.

   LeNet is a simple and small CNN with fewer parameters, so it works well for basic tasks such as handwritten digit recognition. AlexNet is deeper and has many more parameters, allowing it to learn more complex features and achieve better performance on large image datasets. ResNet is even deeper and uses skip connections, which help it train effectively without losing information. Therefore, ResNet generally performs better than AlexNet and LeNet, while LeNet is the simplest and fastest of the three.

21. References

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